

Conjecture 2207: proportion one for a finite fraction field

Local proof package

8 October 2026

Source and interpretation. The source says: “The proportion of 2-generated ideals of a Bezout domain is 1 under finite-quotient-field conditions (2-generated Bezoutization).” We use the usual meaning of *quotient field*: the field of fractions. Here *finite* means finitely many field elements. A two-generated ideal means an ideal of the form (a, b) ; repeated or redundant generators are allowed. In particular $(0, 0) = (0)$ and $(1, 0) = R$ qualify.

For a commutative integral domain R , write $\text{Id}(R)$ for its actual set of ideals and put

$$\text{Id}_2(R) = \{I \in \text{Id}(R) : \text{there are } a, b \in R \text{ with } I = (a, b)\}.$$

When $\text{Id}(R)$ is finite, define its two-generated proportion by $\# \text{Id}_2(R) / \# \text{Id}(R)$. Ideals, rather than presentations by generator pairs, are counted.

Theorem 1. *Let R be a Bézout integral domain and let K be its field of fractions. If K has finitely many elements, then R is a field, $\text{Id}(R)$ consists of exactly the two ideals (0) and R , and*

$$\# \text{Id}_2(R) = \# \text{Id}(R) = 2, \quad \frac{\# \text{Id}_2(R)}{\# \text{Id}(R)} = 1.$$

Proof. The fraction-field map $R \rightarrow K$ is injective. Consequently R is finite; this is a consequence of finite K , not an additional hypothesis. For $a \in R \setminus \{0\}$, the map $x \mapsto ax$ is injective because R is an integral domain. An injective self-map of a finite set is surjective, so some $b \in R$ satisfies $ab = 1$. Thus R is a field.

Let I be an ideal. If I contains a nonzero element a , it also contains $a^{-1}a = 1$, and hence $I = R$. Otherwise $I = (0)$. The ideals (0) and R are distinct because a domain has $0 \neq 1$. They are generated by $(0, 0)$ and $(1, 0)$, respectively. Every ideal is therefore two-generated, the numerator and denominator are both 2, and their quotient is 1. $\square$

Formal correspondence and scope

The Lean file uses the existing mathlib objects `Ideal R`, `IsBezout R`, and `IsFractionRing R K`. The original theorem has precisely the domain, Bézout, fraction-field and finite-field hypotheses:

Ring/domain	<code>[CommRing R]</code> <code>[IsDomain R]</code>
Bézout property	<code>[IsBezout R]</code>
Fraction field	<code>[Field K]</code> <code>[Algebra R K]</code> <code>[IsFractionRing R K]</code>
Finite cardinality of K	<code>[Finite K]</code> .

No `Finite R` or arbitrary `Fintype` object is a premise of the original conclusion. Finiteness of R and of its ideal type is proved from these hypotheses. The real mathlib Bézout class asserts that finitely generated ideals are principal; no replacement axiom is introduced. Its assumption is retained in the final theorem, although the proof establishes the stronger result for every domain with finite fraction field.

Definitions. `TwoGenerated R I` is the proposition

$$\exists a, b \in R, \quad I = \text{span}\{a, b\}.$$

The numerator counts the filter of all actual ideals by this predicate. The denominator counts all actual ideals. The ratio is rational-valued:

Numerator	<code>twoGeneratedCount R</code>
Denominator	<code>Nat.card (Ideal R)</code>
Ratio	<code>twoGeneratedProportion R</code> .

An enumeration of ideals is obtained only after proving their finiteness.

Proof components.

<code>finite_domain_of_finite_fraction_field</code>	Derives finite R .
<code>isField_of_finite_fraction_field</code>	Constructs inverses.
<code>ideal_eq_bot_or_top</code>	Classifies all ideals.
<code>all_ideals_two_generated</code>	Gives actual pairs.
<code>ideal_card_eq_two</code>	Denominator is 2.
<code>twoGeneratedCount_eq_two</code>	Numerator is 2.
<code>original_proportion_one</code>	Proportion is 1.

A concrete $R = K = \mathbb{F}_2$ fraction-field example is also checked.

Interpretation limits. Finite extension degree over a specified base field and finite residue fields are different conditions; neither is supplied in the source or used here. Requiring *minimal* generator number exactly two is also a different convention from being generated by a pair. No asymptotic density or ideal ordering is introduced: the fraction-field hypothesis makes the full set of ideals finite, and the stronger universal generator result applies to it.

Provenance. The complete fraction-field bridge was first checked in this run's scout task and reused in the author package. The source is retained byte-for-byte as `00000002207.md`. Its SHA256 is

0016cb2bf81d535588320e63545be9089d320a430edb18bbe792983b12efb23a.

This document records a local proof package; acceptance and any submission eligibility require the separate manager-owned checks and independent review.